SURESH SIVAKUMAR

Data Engineer

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PROFESSIONAL SUMMARY

Google Cloud - accredited Data Engineer with **4+ years of experience** in building and maintaining **scalable ETL pipelines, real-time data processing systems,** and **cloud-native analytics infrastructure**. Hands-on expertise in optimizing data workflows on GCP (BigQuery, Dataproc, Dataflow, Pub/Sub), Apache Spark, Airflow, SQL, Python and multi-cloud environment (Azure and AWS). Adept in **data modeling, dashboard reporting,** and **cross-functional collaboration.** Seeking to apply advanced **cloud** and **big data expertise** to drive data-driven decisions in fast-paced environments.

PROFESSIONAL EXPERIENCE

ACCENTURE | **Bangalore, India**
Data Engineer | **2021-2024**

- Contributed to design and **implementation of ETL pipelines on GCP** processing **1TB daily data with 99.95% reliability**, optimized **Spark jobs**, improving performance by **60%** while cutting compute costs by **40%** through **transient Dataproc clusters.**
- Collaborated with a **team of 10+ data scientists** to define and implement data requirements, resulting in a **10% increase in model accuracy** for the client's key predictive algorithms.
- Refined workflows for structured data with **schema evolution**, reducing manual effort by **30%.**
- Optimized **Apache Spark** jobs through **partitioning strategies, broadcast joins** and using **transient dataproc,** reducing processing time for 500GB datasets while cutting compute costs.
- Automated ETL workflows using **Apache Airflow,** scheduling twice-daily runs that dynamically create Dataproc clusters with predefined configurations, execute PySpark transformation scripts, and automatically delete clusters post-run to optimize cost efficiency.
- Designed **real-time monitoring dashboards** in **Google Data Studio** with automated alerting, reducing mean time to detection for data quality issues and improving stakeholder decision-making speed by **25%.**

AVP CONSULTING | **Chennai, India**
Junior Developer (Data) | **2020-2021**

- Assisted in building **basic ingestion pipelines** using Azure Data Factory to move data from APIs and SQL sources into Azure Data Lake Storage (ADLS).
- Performed **simple data cleaning and formatting tasks** in Azure Databricks (PySpark), such as handling nulls, filtering records, and preparing tables for reporting.
- Created **Azure Synapse external tables over ADLS** to enable fast analytical queries and Power BI reporting for internal business teams.
- Monitored pipelines using **Azure Monitor, Log Analytics, and Data Factory alerts**, resolving failures related to schema drift, connection issues, and invalid payloads.
- Collaborated closely with senior data engineers to improve **data quality**, implement schema enforcement, and maintain documentation for ingestion workflows.

EDUCATION

MSc in Computer Science(Software Engineering) | **Maynooth, Ireland**
Maynooth University | **2024-2025**

Bachelor of Engineering in Computer Science | **Chennai, India**
Vels Institute of Science, Technology and Advanced Studies | **2016-2020**

TECHNICAL SKILLS

- ***Programming Languages**:* Python, SQL, Bash
- ***Bigdata & Streaming:*** Apache Hadoop, Apache Spark, Apache Kafka, Apache Airflow, Pub, Sub, Cassandra
- ***Cloud Technologies:*** GCP (BigQuery, Dataproc, Dataflow, GCS, Pub/Sub, Composer), AWS (S3, Redshift, EC2), Azure (Data Factory, Synapse), Databricks, Delta Lake, dbt
- ***Visualization Tools:*** Tableau, Google Data Studio

CERTIFICATION

- Google Cloud Certified Professional Data Engineer (Credly Verified)
- Google Cloud Certified Associate Cloud Engineer (Credly Verified)

PROJECTS

Aspect-Based Employee Sentiment Analysis using LLMs (Master's Thesis)

- Designed and implemented an **NLP system** using **BERT, RoBERTa, T5,** and **GPT** variants to extract **aspect-level sentiment** from employee feedback.
- Applied **fine-tuning, multi-task learning**, and **domain adaptation** for improved accuracy.
- Evaluated results using **F1-score** and **human judgment** to validate model performance.

Scalable Data Lakehouse Pipeline – Databricks (Personal Project)

- Designed and implemented a scalable **Data Lakehouse pipeline** using **Databricks Delta Live Tables (DLT)** to process raw data into structured star-schema dimensional and fact tables.
- Ingested raw datasets into **Bronze layer** via **Auto Loader, transformed** and cleansed data into **Silver**, and built **SCD-compliant Dimensions** (Flights, Passengers, Airports).
- Implemented incremental **Change Data Capture (CDC)** logic with surrogate key joins to maintain dimensional relationships.
- Built **Gold fact tables** with **incremental merges** using Delta Lake, enabling reliable analytical queries and reporting.
- **Tools & Tech:** Databricks, PySpark, Delta Live Tables, Delta Lake.